

SENTINEL AI — ALL-IN-ONE SYSTEM DIRECTORY & ROLE MANUAL

Master Technical Guide: Complete Function Directory, Inputs & Outputs, Point-Wise Role Capabilities & Side-by-Side Role Comparison Table

Document Title: Master Directory & Role Capabilities	Classification: RESTRICTED DFIR / SOC
Coverage: 100% Platform APIs & UI Functions	AI Correlation Engine: Google Gemini 1.5 Flash

1. System Architecture & Platform Directory

SENTINEL AI is a unified Digital Forensics & Incident Response (DFIR) and Security Operations Center (SOC) platform. The architecture is designed around live real-time analysis engines: raw RFC-822 email parsing, PCAP frame dissection, static PE/ELF entropy calculation, WHOIS lookups, and SIEM event parsing, all integrated live with Google Gemini AI.

The system strictly segregates capabilities into three roles: **SOC Analyst**, **Forensic Investigator**, and **System Administrator**. Each role operates in a clutter-free, role-gated workspace.

1.1 Database Storage & Live Data Persistence Architecture

- **Embedded Primary Database (SQLite):** Powered by Python standard library sqlite3 using ACID-compliant `forensics_platform.db` with Write-Ahead Logging (`PRAGMA journal_mode=WAL;`) for high concurrency and zero crash data loss.
- **Enterprise Auto-Fallback (MongoDB):** Automatically connects to MongoDB via `pymongo` if an active MongoDB instance is detected at `mongodb://localhost:27017`. If offline, seamlessly switches to embedded SQLite without requiring any user setup.
- **Atomic JSON Backup (`forensics_storage.json`):** Performs atomic double-write snapshots to `forensics_storage.json` on every database write operation to provide human-readable data recovery.
- **Live Data Retention Across Updates:** Startup hooks verify database integrity without purging user data, ensuring 100% of case history, evidence, security audit logs, dashboard metrics, and analyst alert triage statuses survive code updates and server restarts.

2. What Can Be Achieved by Different Roles (Point-Wise Capabilities)

2.1 SOC Analyst Role (Primary Goal: Real-Time Alert Triage & Threat Containment)

- **Rapid Alert Triage:** Evaluate incoming SIEM and EDR security alerts in < 30 seconds. Analysts can dismiss false positives, issue immediate IP containment actions, or escalate critical alerts directly into investigation cases.
- **Custom Threat Detection (YARA & Sigma):** Craft, test, and validate YARA rules and Sigma log detection patterns against live payload text to detect obfuscated PowerShell, Win32 API calls, and C2 traffic.
- **SIEM Log Stream Anomaly Extraction:** Parse raw multi-format log text streams (`.evtx``, Syslog, JSON) to automatically isolate brute-force bursts, shell execution events, and identify top IP talkers.
- **External Attack Surface & CVE Auditing:** Perform external infrastructure health audits to discover open high-risk ports, SSL certificate status, and matching CVSS vulnerabilities on target assets.
- **Instant SOAR Containment Playbooks:** Generate 1-click executable mitigation scripts across Linux (``iptables`` / BIND sinkholes) and Windows PowerShell firewalls to immediately block compromised hosts, IPs, or domains.

2.2 Forensic Investigator Role (Primary Goal: Incident Reconstruction & Case Building)

- ■ **Official Incident Case Management:** Create and maintain formal investigation cases, assign threat severity levels (CRITICAL / HIGH / MEDIUM / LOW), track operational status, and assign case owners.
- ■ **Multi-Vector Artifact Correlation:** Upload and attach email RFC-822 headers, live URL scans, network PCAP packet files, and malware binaries directly to a unified case timeline.
- ■■ **Chronological Attack Reconstruction:** Automatically synthesize multi-file evidence sources into a unified, step-by-step chronological attack timeline.
- ■ **Printable Executive PDF Reports:** Generate professional, printable forensic investigation PDF reports complete with executive summaries, evidence tables, IOC indicators, and remediation blueprints.
- ■ **AI Threat Correlation:** Leverage Google Gemini AI to analyze complex multi-vector evidence files and identify hidden cross-artifact relationships.

2.3 System Administrator Role (Primary Goal: Governance, RBAC & Platform Health)

- ■ **User & Role Management:** Dynamically switch user accounts between `Analyst`, `Investigator`, and `Admin` roles with immediate real-time role updates.
- ■ **AI Engine Key Management:** Configure and persist global Google Gemini API keys across local storage and database backend.
- ■ **Platform-Wide Telemetry & Audit Logs:** View global multi-user metrics, aggregate case totals, and monitor security audit logs across all platform operators.

3. Comprehensive Role Comparison Table

The following side-by-side comparison matrix contrasts the permissions, primary objectives, inputs, outputs, and access boundaries of all three platform roles:

Comparison Dimension	SOC Analyst (`analyst`)	Forensic Investigator (`investigator`)	System Admin (`admin`)
Primary Operational Goal	Rapid alert triage, threat hunting, log parsing & fast containment	Deep-dive case building, multi-evidence correlation & executive reporting	System governance, user role management, AI key config & audit logging
Key Workspace / Route	Analyst Workspace (`/analyst`)	Cases & Incidents (`/cases`)	Admin Control Panel (`/admin`)
Primary Input Data	Triage alerts, raw SIEM log text, YARA/Sigma rules, Target IPs/Domains	Case titles, RFC-822 emails, PCAP files, PE binaries, URL links	Gemini API keys, user role selections, audit log queries
Primary Output Data	Triage statuses, YARA match lines, log anomaly tables, SOAR block scripts	Unified attack timelines, case artifact links, Printable PDF reports	Updated user roles, active AI status, platform audit log stream
Case Management Access	■ Hidden & Restricted (Auto-escalation only)	■ FULL ACCESS (Create, Edit, Delete, Attach)	■ FULL ACCESS (Global view across all users)
Analyst Tools Access	■ FULL ACCESS (Triage, YARA, Logs, SOAR)	■ Hidden & Restricted	■ FULL ACCESS
PDF Report Generation	■ Hidden & Restricted	■ FULL ACCESS (Printable PDF case reports)	■ FULL ACCESS
Admin Control Access	■ Hidden & Restricted	■ Hidden & Restricted	■ FULL ACCESS (Manage users & API keys)

4. Master Platform Function Directory (All Functions & APIs)

Below is the complete function directory detailing inputs, outputs, API routes, and structured return data for all platform tools:

4.1 Email Forensics Engine

POST /api/evidence/parse-email	Route: /email-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload raw `.EML` or `.MSG` email file OR paste raw RFC-822 MIME headers.
INPUT EXAMPLE:	From: attacker@malicious-c2.xyz To: victim@company.com Subject: Urgent Payment Needed Received: from 185.220.101.5
OUTPUT GENERATED:	Extracts hop transit timeline, verifies SPF/DKIM/DMARC authentication, and hashes attachments.
OUTPUT FIELDS:	• hop_trace: Array of IP hops and server hostnames. • auth_results: SPF status, DKIM signature validity, DMARC alignment. • attachment_hashes: MD5 & SHA-256 checksums of extracted email attachments. • threat_level: CRITICAL / HIGH / MEDIUM / LOW rating.

4.2 URL Threat & Infrastructure Analysis Engine

POST /api/evidence/parse-url	Route: /url-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Type or paste any target URL, domain name, or IP address into search box.
INPUT EXAMPLE:	https://login.paypal-security-update.account-verify.xyz/login.php
OUTPUT GENERATED:	Live DNS resolution, WHOIS registrar queries, SSL certificate checks, and brand spoofing detection.
OUTPUT FIELDS:	• resolved_ip: Live server IP address, country, ISP. • domain_age_days: Days since domain WHOIS registration. • shannon_entropy: Randomness rating of URL path. • target_brand_spoof: Identified typosquatting brand target.

4.3 Network PCAP Packet Dissector

POST /api/evidence/parse-pcap	Route: /network-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload binary `.pcap` or `.pcapng` packet capture file (max 50MB).
INPUT EXAMPLE:	binary_capture.pcap (Ethernet/IP frame byte stream)
OUTPUT GENERATED:	Dissects binary frames, reconstructs TCP/UDP streams, and extracts DNS queries.
OUTPUT FIELDS:	• total_packets & total_bytes: Packet count & bandwidth volume. • protocol_breakdown: HTTP, HTTPS, DNS, SSH, FTP traffic ratios. • top_talkers: Source and Destination IPs ordered by packet count. • dns_queries: Array of resolved domain names.

4.4 Malware Static PE Analysis Engine

POST /api/malware/analyze-binary	Route: /malware-forensics Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Upload executable binary file (.exe, .dll, .bin, .elf, .apk, .pdf).
INPUT EXAMPLE:	update_svc.exe (binary Windows Portable Executable)
OUTPUT GENERATED:	Calculates section entropy, extracts Win32 API imports, and flags packing.
OUTPUT FIELDS:	• file_hashes: MD5 & SHA-256 binary checksums. • entropy_score: 0.00–8.00 rating (Entropy > 7.2 flags packing). • suspicious_api_calls: Flagged Win32 APIs (VirtualAllocEx, WriteProcessMemory). • malware_family_hint: Classification hint (Ransomware, Spyware, Trojan).

4.5 Multi-Feed Threat Intelligence Lookup

POST /api/threat-intel/enrich	Route: /threat-intelligence Roles: Analyst, Investigator, Admin
INPUT FORMAT:	Enter any IP address, domain name, file hash (MD5/SHA256), or CVE ID.
INPUT EXAMPLE:	185.220.101.5 OR c5b3a...4f2e
OUTPUT GENERATED:	Queries global threat intelligence feeds for malicious reputation scores.
OUTPUT FIELDS:	• reputation_score: Malicious detection ratio. • known_campaigns: Associated APT groups or malware strains. • asn_owner: Autonomous System Number & hosting organization.

4.6 Analyst SOC Workspace Functions

POST /api/analyst/triage & /test-rule	Route: /analyst Roles: Analyst, Admin Only
INPUT FORMAT:	• Alert Action (FALSE_POSITIVE, CONTAIN, ESCALATE) • YARA / Sigma Rule Code & Test Payload Sample
INPUT EXAMPLE:	<pre>Rule Code: `rule Detect_Powershell { strings: \$s1 = "VirtualAllocEx" condition: \$s1 }` Payload: `powershell.exe -enc ... VirtualAllocEx`</pre>
OUTPUT GENERATED:	Updates alert status (or escalates to Case), and evaluates YARA/Sigma signature matching.
OUTPUT FIELDS:	• triage_status: New alert status (FALSE_POSITIVE, CONTAINED, ESCALATED_TO_CASE). • hunting_result: Match Status (MATCH_FOUND / NO_MATCH), match count, matched line snippets.

4.7 SOAR Playbook Generator

POST /api/analyst/playbooks/generate	Route: /analyst Roles: Analyst, Admin Only
INPUT FORMAT:	Select Action Type ('BLOCK_IP', 'BLOCK_DOMAIN', 'ISOLATE_HOST') and enter Target Value.
INPUT EXAMPLE:	Action: BLOCK_IP Target: 185.220.101.5
OUTPUT GENERATED:	Generates 1-click executable block scripts for Linux and Windows systems.
OUTPUT FIELDS:	• bash_payload: Executable Linux 'iptables' / BIND sinkhole commands. • powershell_payload: Executable Windows PowerShell firewall commands.

4.8 Case Management & Admin Governance Functions

POST /api/cases & PUT /api/admin/users/{id}/role	Route: /cases & /admin Roles: Investigator, Admin
INPUT FORMAT:	• Case Title, Description, Category, Priority Level • Target User ID & New Role ('analyst', 'investigator', 'admin')
INPUT EXAMPLE:	Case: Title='Mail Intrusion', Priority='CRITICAL' Admin Update: UserID='user@company.com', Role='analyst'
OUTPUT GENERATED:	Creates official investigation case, or dynamically updates user role permissions.
OUTPUT FIELDS:	• case_number: Unique Case ID (e.g., CASE-2026-9041). • user_role_update: Updated user permissions and role assignment.

5. Platform File Directory & Architecture Map

The platform backend and frontend code structure is mapped below for reference:

- **Backend Core:** backend/main.py (FastAPI app & CORS setup), backend/app/database.py (JSON / MongoDB store)
- **Authentication & RBAC:** backend/app/auth.py, backend/app/routers/auth_router.py
- **Analyst Workspace APIs:** backend/app/routers/analyst_router.py ('/api/analyst/alerts', '/test-rule', '/parse-logs', '/scan-attack-surface', '/playbooks/generate')
- **Case Management APIs:** backend/app/routers/cases_router.py ('/api/cases'), backend/app/routers/evidence_router.py
- **Forensics Engines:** backend/app/routers/malware_router.py, backend/app/routers/threat_intel_router.py, backend/app/routers/gemini_router.py
- **Frontend Views:** frontend/src/pages/AnalystPage.jsx, Cases.jsx, CaseDetails.jsx, AdminPanel.jsx, Dashboard.jsx
- **Navigation & RBAC Guards:** frontend/src/components/Sidebar.jsx, MobileBottomNav.jsx, Navbar.jsx, App.jsx

SENTINEL AI — DIGITAL FORENSICS & SOC PLATFORM

Comprehensive Technical Guide: Role Functions, Exact Inputs, Expected Outputs & Operational Architecture

Document Version: 2.4.0	Classification: RESTRICTED DFIR
Target Audience: SOC Analysts, Forensic Investigators, Admins	AI Engine: Google Gemini 1.5 Flash

1. System Architecture & Operational Overview

SENTINEL AI is an enterprise-grade Digital Forensics and Incident Response (DFIR) platform combined with a Security Operations Center (SOC) Workspace. The system parses multi-vector evidence streams in real time—including RFC-822 emails, network PCAP binaries, PE/ELF executables, domain WHOIS records, and SIEM event streams—correlated live with the **Google Gemini AI Engine**.

The architecture enforces strict **Role-Based Access Control (RBAC)** so that each security professional operates in a streamlined, uncluttered environment tailored to their exact workflow. All computations run on live dynamic execution engines without pre-packaged static fallbacks.

2. Role-Based Access Control (RBAC) Matrix

The platform segregates duties into three distinct operational roles. Below is the functional permission and visibility matrix:

Feature / Module	SOC Analyst	Forensic Investigator	System Admin
Analyst Workspace (/analyst) (Triage, YARA, SIEM Parser, SOAR)	FULL ACCESS	Hidden & Restricted	FULL ACCESS
Case Management (/cases) (Case Creation, Evidence Attachment)	Hidden & Restricted	FULL ACCESS	FULL ACCESS
PDF Investigation Reports (/reports)	Hidden & Restricted	FULL ACCESS	FULL ACCESS
Standalone Forensics Tools (Email, URL, PCAP, Malware, Threat Intel)	FULL ACCESS	FULL ACCESS	FULL ACCESS

Gemini AI Chat Assistant	FULL ACCESS	FULL ACCESS	FULL ACCESS
Admin Control Panel (/admin)	Hidden & Restricted	Hidden & Restricted	FULL ACCESS

3. Complete Module Input & Output Specifications

This section details the exact input required for every tool in each role, alongside the precise structured output generated by the engine.

3.1 Email Forensics Module (/email-forensics)

Email RFC-822 Header & Attachment Analyzer	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload raw <code>`EML`</code> or <code>`MSG`</code> email file or paste raw RFC-822 MIME headers into text input area.
INPUT EXAMPLE:	From: CEO <ceo@company.com> To: Finance <finance@company.com> Subject: Urgent Wire Transfer Received: from mail.attacker.com (185.220.101.5)
GENERATED OUTPUT SUMMARY:	Parses email headers, verifies authentication signatures, computes attachment checksums, and assesses phishing risk.
STRUCTURED OUTPUT FIELDS:	• Hop Trace: Array of IP hops, timestamps, and server hostnames. • Auth Checks: SPF (Pass/Fail), DKIM signature validity, DMARC alignment. • Attachment Hashes: MD5 & SHA256 cryptographic hashes of attachments. • Risk Score: 0–100 Threat Rating (CRITICAL / HIGH / MEDIUM / LOW).

3.2 URL Threat Analysis Module (/url-forensics)

Live URL & Domain Infrastructure Analyzer	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Type or paste any target URL, domain name, or IP address into the search input box.
INPUT EXAMPLE:	https://login.paypal-security-update.account-verify.xyz/login.php
GENERATED OUTPUT SUMMARY:	Executes live DNS resolution, WHOIS registrar queries, SSL certificate verification, and brand spoofing detection.
STRUCTURED OUTPUT FIELDS:	• Resolved IP & Location: Server IP address, country, ISP, and hosting provider. • Domain Age: Days since registration (flags domains < 30 days old). • Shannon Entropy: Mathematical randomness score of URL path. • Brand Spoofing Flag: Identifies typosquatting target (e.g. PayPal, Google, Microsoft).

3.3 Network PCAP Forensics Module (/network-forensics)

Binary PCAP Network Stream Dissector	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload binary <code>.pcap</code> or <code>.pcapng</code> packet capture file (max 50MB per upload).
INPUT EXAMPLE:	<code>sample_capture.pcap</code> (binary Ethernet/IP frames containing TCP/UDP/DNS flows)
GENERATED OUTPUT SUMMARY:	Dissects packet layers, reconstructs TCP conversations, flags beaconing IPs, and extracts DNS domain queries.
STRUCTURED OUTPUT FIELDS:	• Bandwidth & Packet Metrics: Total packets, total bytes, duration. • Protocol Breakdown: HTTP, HTTPS, DNS, SSH, FTP packet ratios. • Top IP Talkers: Source/Destination IPs ordered by packet volume. • Extracted DNS Queries: List of resolved domain names from DNS frames.

3.4 Malware Forensics Module (/malware-forensics)

PE / ELF Static Binary Analysis Engine	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Upload executable binary file (.exe, .dll, .bin, .elf, .apk, .pdf).
INPUT EXAMPLE:	update_svc.exe (binary Windows Portable Executable file)
GENERATED OUTPUT SUMMARY:	Computes entropy across file sections, extracts Win32 API import flags, and identifies malware packing.
STRUCTURED OUTPUT FIELDS:	• File Hashes: MD5 & SHA-256 binary checksums. • Entropy Score: 0.00–8.00 rating (Entropy > 7.2 flags packing/encryption). • Suspicious API Imports: Flags APIs like VirtualAllocEx, WriteProcessMemory, CreateRemoteThread. • Malware Family Hint: Heuristic classification (e.g., Ransomware, Spyware, Trojan).

3.5 Threat Intelligence Module (/threat-intelligence)

Multi-Feed Threat Intel IOC Lookup	ROLE: Analyst, Investigator, Admin
REQUIRED INPUT FORMAT:	Enter any IP address, domain name, file hash (MD5/SHA256), or CVE ID.
INPUT EXAMPLE:	185.220.101.5 OR c5b3a...4f2e OR attacker-c2.xyz
GENERATED OUTPUT SUMMARY:	Queries multi-feed threat intelligence databases for reputation scores and campaign tagging.
STRUCTURED OUTPUT FIELDS:	• Reputation Score: Malicious detection ratio across threat feeds. • Known Campaigns: Associated APT groups or malware operations. • Geolocation & ASN: Autonomous System Number and host organization.

3.6 Analyst SOC Workspace (/analyst)

Dedicated workspace for Security Operations Center Analysts containing 5 interactive sub-tools:

Sub-Tool	Input Required	Output Generated
1. Alert Triage Queue	Click action buttons: FALSE POSITIVE, CONTAIN IP, or ESCALATE on incoming alerts.	Alert status changes immediately. ESCALATE automatically creates an official investigation case.
2. Threat Hunting Workbench	• Select YARA or SIGMA mode. • Paste Rule Code in textarea. • Paste Test Log/Payload sample.	Signature validation report: Match Status (MATCH_FOUND / NO_MATCH), match count, and matched line snippets.
3. SIEM Log Parser	Paste raw Syslog, EVTX, or JSON log text stream into log input box.	Structured anomaly summary: Lines parsed, Failed login count, Shell execution events, and Top IP talkers.

4. Attack Surface Scanner	Type target IP address or Domain name into search box.	Security Health Grade (A–F), overall risk score, open ports table, and matching CVE vulnerabilities.
5. SOAR Playbook Engine	Select action type (Block IP, Sinkhole Domain, Isolate Endpoint) and enter Target Value.	Executable 1-click containment payloads: Linux iptables / BIND scripts and Windows PowerShell firewalls.

3.7 Case Management & Incident Reports (/cases & /reports)

Incident Case Creator & PDF Report Generator	ROLE: Investigator, Admin
REQUIRED INPUT FORMAT:	- Case Title, Description, Category, Threat Level (CRITICAL / HIGH / MEDIUM / LOW). - Upload evidence artifacts to case.
INPUT EXAMPLE:	Title: 'Corporate Mail Server Ransomware Intrusion' Threat Level: CRITICAL Category: Malware
GENERATED OUTPUT SUMMARY:	Unified case file, automated attack timeline, evidence correlation matrix, and downloadable executive PDF report.
STRUCTURED OUTPUT FIELDS:	Case Number: Unique ID (e.g. CASE-2026-9041). Unified Timeline: Chronological sequence of all uploaded artifacts. Executive Report PDF: Printable PDF containing executive summary, IOC table, and remediation steps.

4. Step-by-Step Operator Workflow Guide

Step 1: Logging In & Role Verification

Navigate to /login and log in with your account credentials. Check the bottom of the left sidebar to verify your assigned role (**ANALYST**, **INVESTIGATOR**, or **ADMIN**).

Step 2: Analyst Triage Workflow (For SOC Analysts)

- Go to **Analyst Workspace** (/analyst).
- In the **Alert Triage Queue**, review incoming alerts. Click FALSE POSITIVE for noise, or CONTAIN IP to generate block scripts.
- Click ESCALATE on high-severity alerts to create an official case.
- Use the **SIEM Log Parser** or **Threat Hunting Workbench** to validate YARA/Sigma detection rules against new payloads.

Step 3: Investigation Workflow (For Forensic Investigators)

- Go to **Cases & Incidents** (/cases).
- Click **NEW INVESTIGATION CASE** and fill in incident details.
- Open the case details and upload evidence into Email, URL, PCAP, or Malware tabs.
- Click **GENERATE PDF REPORT** to export an official forensic report.

Step 4: AI Correlation & System Administration (For Admins)

1. Configure your Google Gemini API key in **Admin Control Panel** (/admin).
2. Monitor platform audit trail logs and switch user roles as needed.